µPhon Stereo Headphone Amplifier Module

Rev. 1

Description

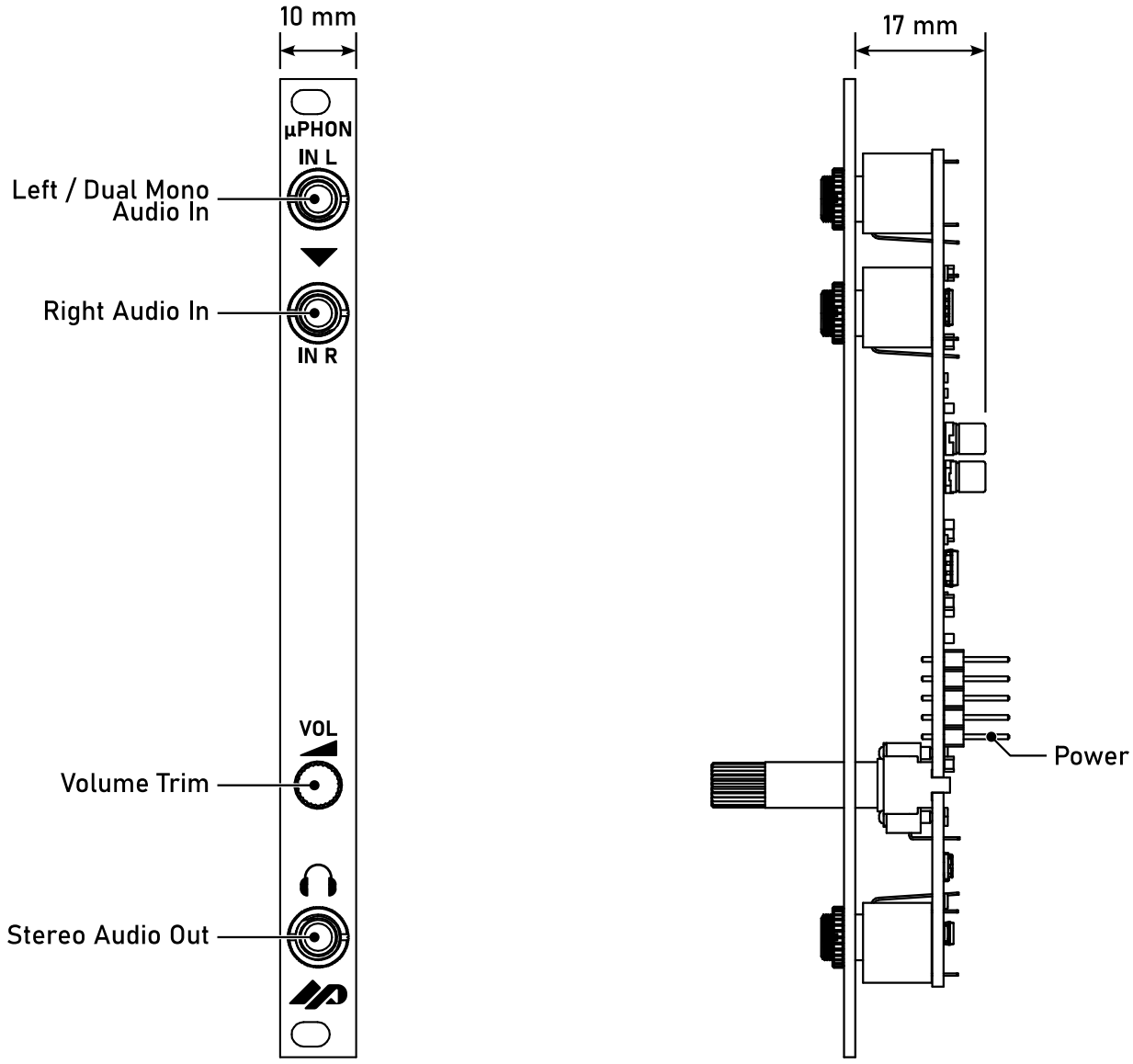
The µPhon is a low-noise stereo/dual mono headphone amplifier for Eurorack systems in a compact 2hp form factor. The output can drive a wide variety of headphones with various driver impedances due to its high 100mA current output capability per channel. Signal integrity is maintained using low-distortion, low-noise amplifiers.

Features

- Outputs driven by low noise, high current OPA1656 design up to 100mA
- SSI2162 volume gain control
- Normalled inputs for stereo or dual-mono operation
- Compact 2hp, 17mm deep design
- Power input reverse-polarity protection
- Current Draw*: +12V: 20mA, -12V: 14mA

*Using 300Ω headphones. May vary depending on headphone impedance.

Module Layout



Functionality

Inputs

The left audio input jack is normalled to the right channel to allow both stereo and dual-mono operation.

Both inputs are AC-coupled to protect headphones from DC current, and can safely tolerate signals up to +/- 12V.

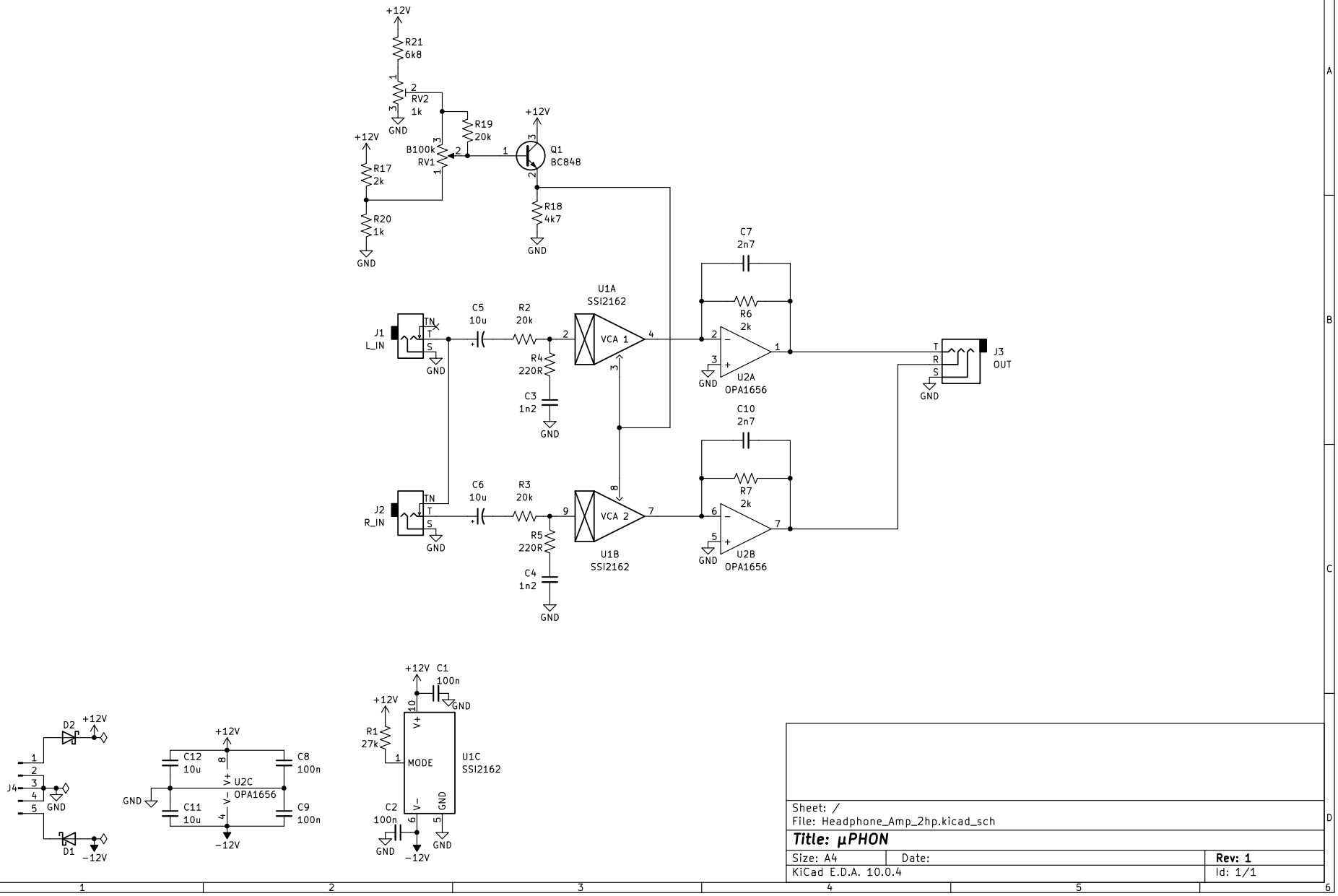
Volume Control

Output volume for both channels is controlled via a single potentiometer that drives a VCA gain cell on each channel.

Maximum output volume is set with the trim potentiometer on the back side of the module labeled MAX VOL. This should be set to the minimum volume (fully counter-clockwise) before using the module. Slowly adjust the potentiometer to set to a comfortable listening level.

Bill of Materials

Value	Qty.	Name on Board	Description	Footprint
220Ω	2	R4, R5	220Ω 1% 0.125W Resistor	0805
1kΩ	1	R20	1kΩ 1% 0.125W Resistor	0805
2kΩ	3	R6, R7, R17	2kΩ 1% 0.125W Resistor	0805
4.7kΩ	1	R18	4.7kΩ 1% 0.125W Resistor	0805
6.8kΩ	1	R21	6.8kΩ 1% 0.125W Resistor	0805
20kΩ	3	R2, R3, R19	20kΩ 1% 0.125W Resistor	0805
27kΩ	1	R1	27kΩ 1% 0.125W Resistor	0805
1.2nF	2	C3, C4	1.2nF 100V C0G MLCC	0805
2.7nF	2	C7, C10	2.7nF 50V C0G MLCC	0805
100nF	4	C1, C2, C8, C9	100nF 50V X7R MLCC	0805
10uF	2	C11, C12	10uF 25V X5R MLCC	0805
10uF	2	C5, C6	10uF 25V Elec. Capacitor	4x5.7mm SMD
Schottky Diode	2	D1, D2	Schottky Diode 40V	SOD-123
BC846 NPN	1	Q1	BC846 NPN Transistor	SOT-23
SSI2162	1	U1	SSI2162 Dual VCA	SSOP-10 50mil
OPA1656	1	U2	OPA1656 Dual Op Amp	SOIC-8 150mil
B100k Pot	1	RV1	100k Lin. Potentiometer	Alpha 9mm vertical
B1k Pot	1	RV2	1k Lin. Potentiometer	Bourns TC33X vertical
3.5mm Jack	1	J1, J2	3.5mm Mono Jack	PJ-3001F
3.5mm Jack	1	J3	3.5mm Stereo Jack	PJ-3001F
5-Pin Header	1	J4	2.54mm Header 5x1	2.54mm 5-pin right angle



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